

Name: _____



Lowest common multiple

Task A: Common multiples using prime factorisation

1) Write the first eight multiples of the following:

a) 3×5

b) $2^2 \times 3$

c) $2 \times 3^2 \times 5$

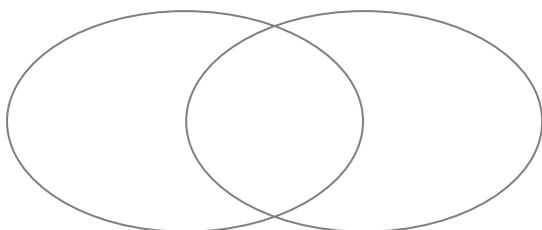
d) What is the first common multiple of 15 and 12?

Task B: Lowest common multiple using prime factorisation

1) Find the **LCM** of the following pairs of numbers:

a) $56 = 2 \times 2 \times 2 \times 7$

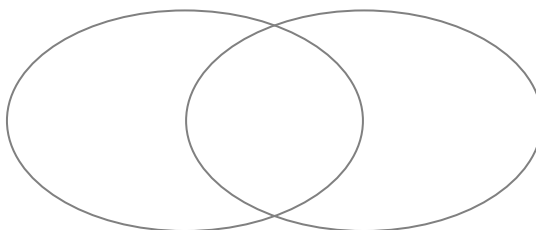
$140 = 2 \times 2 \times 5 \times 7$



The **LCM** of 56 and 140 is

b) $84 = 2 \times 2 \times 3 \times 7$

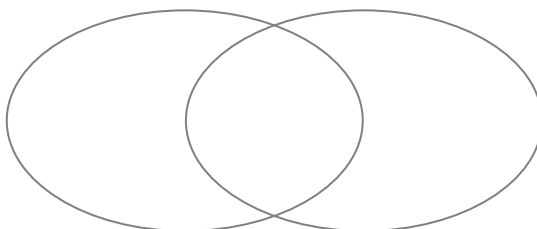
$70 = 2 \times 5 \times 7$



The **LCM** of 84 and 70 is

c) $2016 = 2^5 \times 3^2 \times 7$

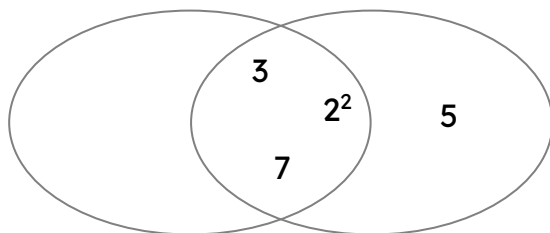
$1512 = 2^3 \times 3^3 \times 7$



The **LCM** of 2016 and 1512 is

2) The **LCM** of the 420 and another number is represented in the Venn diagram is 1260.

- What is the missing prime factor?
- What was the other number?



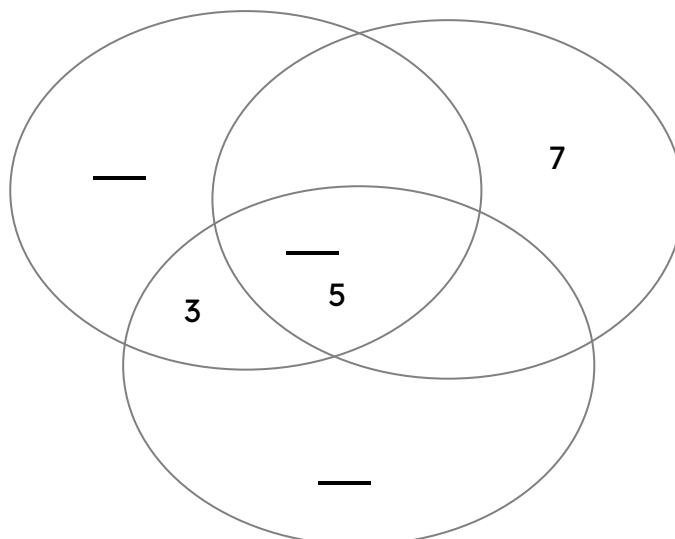
Task C: Lowest common multiple of more than two numbers

Fill in the blanks.

$$180 = 2^2 \times \underline{\quad} \times 5$$

$$\underline{\quad} = 2^2 \times 5 \times 7$$

$$\underline{\quad} = 2^2 \times 3 \times 5 \times 11$$



The **LCM** of 180, $\underline{\quad}$ and $\underline{\quad}$ is $\underline{\quad}$